

Hongshan Chen

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Availability: Full-time, May 2027 – Jan 2028

EDUCATION

National University of Singapore

Aug 2024 – Jul 2028

Bachelor of Computing in Computer Science, minoring in Interactive Media

Singapore

- **Awards:** NUS Science and Technology Undergraduate Scholarship | **GPA:** 4.30/5.00
- **Teaching Assistant:** Computer Graphics (AY26/27 Sem 1); Software Engineering (reserve tutor)

EXPERIENCE

Algorithm Intern | LLM Agent Evaluation, Hotel R&D

Jun 2026 – Aug 2026

Trip.com Group (Ctrip)

Shanghai, China

- Independently owned an **end-to-end LLM travel-agent evaluation system** — corpus construction, question generation, answer keys grounded in mechanically decidable attributes, automated answer-quality scoring, and leaderboards — built to produce rankings that survive re-runs, addressing the non-reproducibility of LLM-as-judge grading.
- Derived a **query intent taxonomy** (room type, restaurants, cancellation policy, ...) from real user discussions via open coding with entropy-based saturation analysis, then applied engagement-weighted sampling to align benchmark coverage with the observed query distribution.
- Validated reliability through **sample-size and ranking-stability analysis** (130 questions $\times$ 5 models $\times$ 3 runs = 1,950 evaluations), establishing how many items are needed before the leaderboard ordering stops moving.
- Engineered a **concurrent Python pipeline** (async workers, Playwright, cascaded LLM filtering) that crawls, deduplicates, and validates real user discussions into a structured retrieval corpus.
- Preparing a **first-author manuscript**, with submission planned for late 2026.

Undergraduate Researcher | NUS Odyssey Research Programme

May 2026 – Present

National University of Singapore

Singapore

- Co-designed a **long-horizon evaluation benchmark for video world models**, testing whether generated videos preserve evidence and maintain a consistent 3D world across camera departure, occlusion, viewpoint changes, and revisits; manuscript in preparation for submission.
- Generated controlled synthetic data in **Unreal Engine 5** (scripted camera trajectories, revisit paths, per-object segmentation) — chosen over in-the-wild video so that failures are attributable to the model rather than to uncontrolled scene variation.
- Contributed to designing **quantitative metrics and evaluation protocols** for reproducibly measuring these failure modes across models.

PROJECTS

Harmonia | TikTok TechJam 2025 | FAISS, RAG, CLIP, vLLM [\[Project Link\]](#)

Aug 2025 – Sep 2025

- Led architecture of a **4-stage cascaded multimodal pipeline** for four-way review-quality classification (valid vs. three violation types) on Google location reviews: rule-based filtering $\rightarrow$ FastBERT with entropy-based confidence routing $\rightarrow$ Qwen3-8B + CLIP text-image alignment $\rightarrow$ RAG verification over a FAISS index, served via vLLM — routing cheap traffic to cheap models and escalating only on low-confidence samples.
- Profiled the **accuracy-latency frontier**: $\sim 92\%$ accuracy at $\sim 150\text{ms/sample}$, roughly 5 points above the latency-matched blend of the two endpoint models — evidence that confidence routing beats naive mixing.

NetEase Cloud Music Support Agent | Spring Boot, RAG, Redis [\[Project Link\]](#)

Dec 2025 – Jan 2026

- Built a **RAG support-agent backend** in Spring Boot with top-K retrieval and query normalisation; added a no-evidence refusal gate that short-circuits generation when retrieval returns zero hits, trading answer coverage for groundedness so the agent declines rather than fabricates.
- Wrote a load-testing harness (500 requests @ concurrency 20) that **diagnosed a JDBC connection-pool ceiling**, raising throughput $3.5 \rightarrow 8.3$ req/s; added retry/circuit-breaker, idempotent writes, and Prometheus metrics.

TECHNICAL SKILLS

AI & ML: offline evaluation & metric design, statistical experiment design, information retrieval (vector search, RAG, retrieval corpus construction), LLM agents (planning, tool use), multimodal learning (CLIP), video understanding, prompt engineering

Programming: Python, C++, Java, C, SQL, JavaScript

Tools: PyTorch, FAISS, vLLM, NumPy, Spring Boot, Redis, MySQL, Docker, Playwright, async Python, Git/GitHub Actions, Linux/CLI, Unreal Engine 5

LEADERSHIP & ACTIVITIES

Logistics Head, SoC Freshmen Finale Camp 2026 — led logistics planning, vendor coordination, and budget tracking.

NUS Piano Ensemble — annual ensemble concerts. | **NUS Climbing Club** — SNCS Level 2 Sport Climbing certification.

Last updated: Aug 2026